Breast Ultrasound Image Classification Using Transfer Learning

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Abstract

Breast cancer is one of the most common causes of cancer-related deaths among women worldwide. Early detection significantly improves treatment outcomes. Ultrasound imaging is widely used for breast cancer diagnosis because it is non-invasive and relatively inexpensive.

This project presents a deep learning approach for classifying breast ultrasound images from the BUSI dataset into three categories: Benign, Malignant, and Normal. Transfer learning using a pre-trained ResNet50 convolutional neural network is applied to extract high-level image features. Data augmentation and preprocessing techniques were used to improve model generalization. The trained model was evaluated using accuracy, confusion matrix, and classification metrics.

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1 Introduction

Breast cancer remains a major health concern worldwide. Early diagnosis plays a crucial role in improving survival rates.

Medical imaging techniques such as ultrasound are commonly used for detecting breast abnormalities. However, manual interpretation of ultrasound images requires experienced radiologists and may lead to subjective variations.

Recent advances in deep learning, particularly convolutional neural networks (CNNs), have shown promising results in medical image analysis. Transfer learning allows models trained on large datasets such as ImageNet to be adapted for smaller medical datasets.

The objective of this project is to build an automated system for classifying breast ultrasound images into three classes:

- Benign
- Malignant
- Normal

2 Dataset Description

The dataset used in this project is the BUSI (Breast Ultrasound Images) dataset. It contains ultrasound images collected from female patients.

The dataset contains three categories:

- Benign
- Malignant
- Normal

Class	Number of Images	Percentage
Benign	437	56%
Malignant	210	26.9%
Normal	133	17.1%

Table 1: Class distribution in BUSI dataset

Mask images were removed during preprocessing and only original ultrasound images were used for model training.

3 Methodology

3.1 Data Preprocessing

The preprocessing pipeline included the following steps:

- Removing segmentation mask images
- Resizing images to 224×224
- Applying ResNet50 preprocessing
- Normalizing pixel values

3.2 Data Augmentation

To increase the diversity of training data, several augmentation techniques were applied:

- Image rotation
- Horizontal flipping
- Zoom transformation
- Random shifts

These transformations help the model generalize better to unseen images.

3.3 Model Architecture

A transfer learning approach using the pre-trained ResNet50 model was used.

The architecture consists of:

- Pre-trained ResNet50 base model
- Global Average Pooling layer
- Dense layer with ReLU activation
- Dropout layer
- Softmax output layer for three classes

3.4 Training Strategy

Training was performed in two stages.

Phase 1: The base model layers were frozen and only the classification layers were trained.

Phase 2: The last 30 layers of the base model were unfrozen and fine-tuned with a smaller learning rate.

4 Results

4.1 Training Accuracy

Figure 1 shows the training and validation accuracy across epochs.

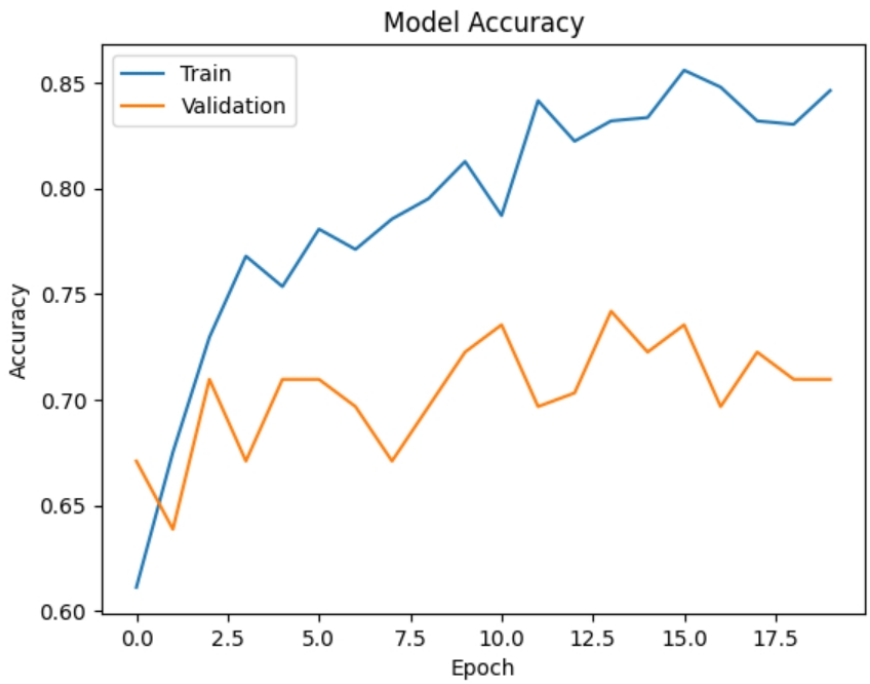

Figure 1: Training and validation accuracy curve

The model achieved approximately:

- Training Accuracy: $\sim 85\%$
- Validation Accuracy: $\sim 71\%$

4.2 Confusion Matrix

The confusion matrix illustrates the distribution of predicted and actual labels for the validation dataset.

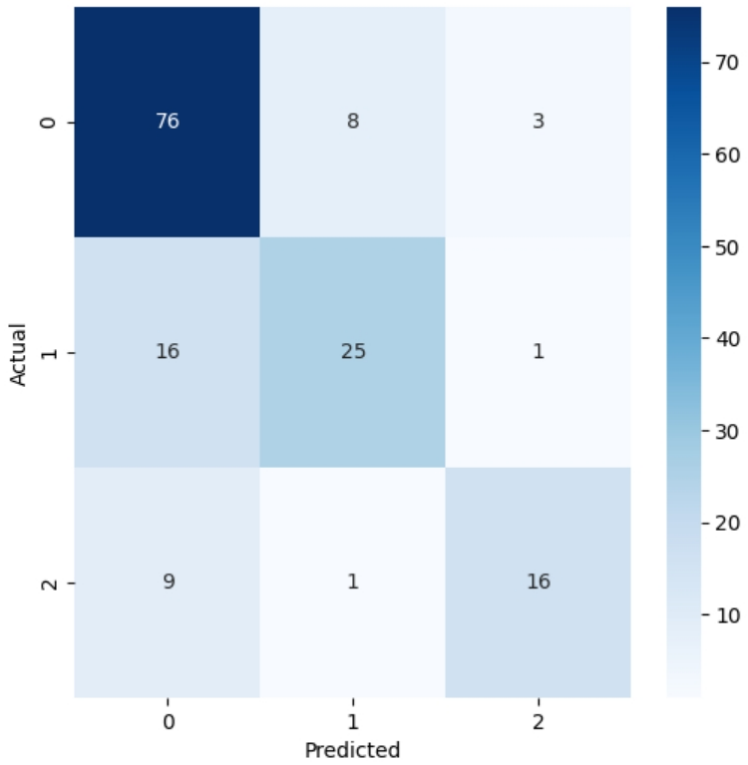

Figure 2: Confusion matrix for BUSI classification

4.3 Classification Metrics

The model performance was evaluated using the following metrics:

- Precision
- Recall
- F1-score

These metrics were calculated using the classification report generated from the trained model.

5 Discussion

The experimental results demonstrate that transfer learning is effective for medical image classification tasks.

Using ResNet50 allows the model to leverage feature representations learned from large datasets such as ImageNet. Data augmentation also helped improve generalization.

Although the model achieved good performance, the gap between training and validation accuracy indicates slight overfitting.

Future improvements could include:

- Training with larger medical datasets
- Using ensemble models
- Applying explainable AI techniques

6 Conclusion

This project presented a deep learning framework for breast ultrasound image classification using the BUSI dataset.

Transfer learning with ResNet50 enabled effective feature extraction and classification of ultrasound images into Benign, Malignant, and Normal classes.

The model achieved promising results and demonstrates the potential of deep learning for assisting medical diagnosis.

7 References

- BUSI Breast Ultrasound Dataset
- TensorFlow Documentation
- Transfer Learning for Medical Image Classification